

TABLE 4
Estimated programme impacts
*Monthly average impacts in dollars over 18 months after random assignment**

	Experimental impact $M(S_E)$	Experimental impact for persons in overlap support region $M(S_P)$	Bias from non-overlap	% bias
Adult males	44 (17)	61 (17)	17	39
Adult females	29 (9)	35 (12)	6	21
Male youth	-58 (14)	-36 (18)	22	38
Female youth	-1 (11)	25 (18)	26	2500

* In our data, the experimental control group was administered a long-baseline survey that gathered five years of retrospective data while the experimental treatment group was not. Since information on recent labour force status and on recent earnings is missing for treatments, we are only able to obtain coarse estimates of P for the treated group. We use the coarse II model described in the notes to Table 6(a). The support region in the nonexperimental analysis is determined using the best predictor P model, so it is necessary to estimate which treatment group members would be excluded by imposing a common support to obtain impact estimates using nonexperimental methods. The impact estimates in the support region were obtained as follows. For controls and treatment, we first divide the coarse P distribution into 20 equal-size bins, then within-bin treatment estimates are estimated. The impact estimate in the overlap region is obtained as the weighted average of the within-bin estimates, with the weights given by the proportion of controls within each bin after deleting controls whose values of P lie outside the overlap region.

control group members ($D=0$). The lack of overlap and the restriction on the range of P means that the empirical counterpart to (A-2) is violated in our samples.

Table 4 presents empirical evidence on how the estimated programme effect changes when it is estimated on S_E and on S_P , where S_P is the region of overlapping support of P . The estimated experimental impact on earnings is given in the first column, while the experimental impact for the support of P common to participants and ENP comparison group members is presented in the next column. The bias is positive for all demographic groups and expressed as a fraction of the impact estimated over the full support of the programme is sizeable. (See the final column of the table.)

Nonparametric matching methods can only be meaningfully applied over regions of overlapping support. Simple nearest neighbour matching estimators could be mechanically applied over the whole region; but, as is evident from Figure 1, matches for experimental controls with high P values are likely to be poor. If nearest neighbour matching is performed with replacement, impact estimates are likely to be sensitive to the inclusion or omission of a few persons in the comparison group sample who are used repeatedly as matches for high P controls. Below we show that, in general, nearest neighbour estimators are more reliable in our application when matches are restricted to regions of overlapping support. Nonparametric kernel-based methods require that the density of the matching variables be strictly bounded away from zero. Therefore these methods require that the region of overlapping support be determined.

Appendix C discusses the details of how we impose the common support condition in constructing S_P , and an expanded version of Appendix C, available on request, presents some Monte Carlo evidence on the performance of alternative procedures for selecting S_P . In large samples, the alternative rules we investigate produce the same estimates. This is not true, however, in small samples and our results for male youth are somewhat sensitive to the choice of the procedure used to select S_P .

12.2. *Estimates of bias for alternative matching estimators*

Tables 5(a)–5(b) report estimates of the bias $\tilde{B}_{S_p}$ from a variety of matching estimators for the four demographic groups analysed in this paper. The first column in each table reports the unadjusted mean difference experimental control and nonparticipant earnings, for each quarter and averaged over the 18 month period following the date of randomization or eligibility determination ($t = 1$ to $t = 6$). The second row from the bottom shows the bias as a percentage of the estimated post-programme impact, obtained using experimental data over its full support. The final row of each table reports the bias as a percentage of the experimental impact estimated over the support of the nonexperimental estimator, S_P . We tried “optimal” bandwidths for the kernel estimators but found that they give less smooth estimates.²⁷ A bandwidth of 0.06 is used in this paper. Results are comparable for other fixed bandwidths within ± 0.02 of 0.06 and for the average of the optimal bandwidths. For adult men, the average post-programme bias in the raw means is $-\$337$ per month, which is almost nine times the estimated programme impact for them and seven times the average conditional difference-in-differences estimator of $\$52$.

The nearest neighbour estimator that does not impose common support does surprisingly well for adult men, but this is not true for the other groups where imposing the common support restriction leads to a substantial reduction in bias. In results available on request, we show that simple averaging over the 5 or 10 closest neighbours usually improves its performance, although for adult women the bias is somewhat higher with the averaged estimator. P score matching estimators that use local linear regression weights, shown in the next two columns of the table usually perform roughly the same as the averaged nearest neighbour estimators. In results available on request from the authors, we obtain comparable results for smoothed and unsmoothed Mahalanobis metric estimators. The local linear regression estimator performs slightly better for youth groups but has no edge over the other matching estimators for the adult groups. The conditional difference-in-differences estimator performs somewhat better than the matching estimators for some groups but it is by no means the dominant estimator, despite the fact that the estimator is the only one for which the identifying assumptions are not rejected by formal statistical tests. Elsewhere (Heckman, Ichimura, Smith and Todd (1996a, b)), we document that the relatively strong performance of the matching estimators in our samples is a result of offsetting biases. The pointwise bias $B(P)$ is not zero and is substantially negative for low values of P . For different intervals of P , the bias in the matching estimators is substantial.²⁸ However, the pointwise bias $B(P)$ for the difference-in-differences estimator is much closer to zero.

Although the estimated biases are small relative to the simple difference in means, they are still a substantial fraction of total programme impact whether expressed over the full support of the experiment or the restricted support from the nonexperimental evaluation samples. (See the bottom two rows.) Moreover, there is substantial quarter-to-quarter variation in the estimated bias.

13. THE IMPORTANCE OF THE CONDITIONING VARIABLES

We next consider the effect of reducing the information used to predict programme participation. Previous nonexperimental evaluations of training programmes had access to much

27. The optimal plug-in bandwidth is derived in Heckman, Ichimura, Smith and Todd (1996c).

28. Persons with low P who participate in the programme have substantially lower earnings than nonparticipants.

TABLE 5(a)

Estimated bias for alternative nonparametric matching methods
Experimental controls and eligible nonparticipants (ENPs)†‡*

Quarter	Difference in means (β)	Nearest neighbour without common support	Nearest neighbour with common support	Local linear P score matching	Regression-adjusted local linear matching	Difference-in-differences from local linear P score matching	Difference-in-differences from regression-adjusted local linear matching
Adult males							
$t=1$	-418 (38)	221 (56)	123 (67)	33 (59)	39 (60)	97 (62)	104 (63)
$t=2$	-349 (47)	-166 (151)	77 (83)	37 (61)	39 (64)	77 (89)	77 (92)
$t=3$	-337 (55)	-58 (206)	53 (96)	29 (78)	21 (80)	90 (114)	74 (114)
$t=4$	-286 (57)	161 (178)	86 (96)	80 (77)	65 (82)	112 (90)	98 (91)
$t=5$	-305 (57)	167 (196)	87 (100)	64 (77)	50 (83)	19 (95)	-5 (99)
$t=6$	-328 (63)	45 (191)	34 (113)	37 (82)	17 (90)	4 (105)	-35 (111)
Ave. 1 to 6	-337 (47)	62 (127)	77 (80)	47 (60)	38 (64)	67 (71)	52 (74)
As a % of impact**	775%	142%	177%	108%	87%	153%	120%
As a % of adjusted impact	552%	102%	126%	77%	62%	109%	85%
Adult females							
$t=1$	-26 (24)	115 (30)	67 (36)	45 (33)	55 (36)	65 (31)	74 (30)
$t=2$	29 (25)	113 (53)	47 (46)	48 (37)	55 (39)	53 (40)	60 (39)
$t=3$	38 (26)	124 (107)	63 (59)	26 (48)	31 (52)	10 (56)	14 (59)
$t=4$	55 (30)	106 (102)	58 (52)	36 (39)	35 (45)	12 (53)	7 (56)
$t=5$	62 (34)	92 (111)	47 (51)	48 (40)	48 (45)	29 (51)	23 (53)
$t=6$	40 (36)	79 (84)	-6 (54)	23 (40)	16 (42)	-5 (51)	-18 (51)
Ave. 1 to 6	33 (26)	105 (69)	46 (43)	38 (33)	40 (38)	27 (38)	27 (39)
As a % of impact**	113%	358%	157%	130%	137%	93%	91%
As a % of adjusted impact	94%	300%	131%	109%	114%	78%	76%

* The table reports the bias for alternative matching methods. The bias in the first column is $\hat{\beta}$. The estimator in the second column does not restrict matches to a common support region. The estimators in the third through seventh columns restrict matches to a common support region and the bias estimates correspond to B_{SE} .

† The best predictor model given in the second footnote to Table 2, is used to estimate the probability of programme participation. The conditioning variables in the regression adjusted local linear models are site, race, age, education, previous training, work experience in months, the local unemployment rate, indicator variables for marital status and for the presence of a child aged less than 6 in the household, and indicators for the quarter of the year and the year.

‡ A 2% trimming rule is used to determine the region of overlapping support (see Appendix C). A fixed bandwidth equal to 0.06 and a biweight kernel, defined in Appendix A, are used for the nonparametric estimates.

** The impacts in the table are mean monthly impacts for the six post-programme quarters, estimated using the experimental treatment and control data for the four JTPA training sites in our study. See the experimental impacts and the adjusted impacts in Table 4.

TABLE 5(b)

Estimated bias for alternative nonparametric matching methods
Experimental controls and eligible nonparticipants (ENPs)†‡*

Quarter	Difference in means ($\hat{\beta}$)	Nearest neighbour without common support	Nearest neighbour with common support	Local linear P score matching	Regression-adjusted local linear matching	Differences in local linear P score matching	Differences in regression-adjusted local linear matching
Male youth							
$t=1$	-51 (58)	146 (92)	49 (75)	3 (64)	8 (61)	43 (72)	80 (77)
$t=2$	2 (60)	197 (92)	98 (82)	40 (64)	28 (55)	43 (60)	61 (60)
$t=3$	5 (73)	202 (105)	83 (119)	33 (81)	-8 (77)	92 (80)	70 (86)
$t=4$	17 (69)	246 (105)	98 (94)	44 (81)	4 (71)	9 (74)	-5 (77)
$t=5$	82 (73)	283 (118)	138 (89)	84 (93)	42 (76)	18 (88)	-11 (81)
$t=6$	65 (77)	258 (145)	129 (121)	28 (93)	-31 (92)	-23 (89)	-64 (84)
Ave. 1 to 6	20 (57)	222 (88)	99 (78)	39 (66)	7 (53)	30 (49)	22 (48)
As a % of impact**	34%	382%	170%	67%	12%	52%	38%
As a % of adjusted impact	56%	617%	275%	108%	19%	84%	61%
Female youth							
$t=1$	6 (31)	67 (54)	-7 (60)	31 (42)	-8 (46)	-7 (38)	-14 (41)
$t=2$	54 (40)	85 (57)	23 (60)	79 (53)	27 (49)	60 (49)	27 (47)
$t=3$	89 (44)	142 (62)	97 (78)	121 (60)	49 (52)	135 (59)	83 (58)
$t=4$	42 (50)	89 (56)	24 (72)	37 (59)	-28 (59)	45 (57)	4 (59)
$t=5$	64 (41)	121 (57)	51 (63)	65 (54)	8 (54)	45 (61)	-7 (63)
$t=6$	31 (46)	107 (82)	34 (70)	34 (65)	1 (62)	31 (70)	6 (69)
Ave. 1 to 6	48 (36)	102 (49)	37 (56)	61 (45)	8 (42)	52 (39)	17 (39)
As a % of impact**	7059%	15000%	5441%	8971%	1176%	7574%	2426%
As a % of adjusted impact	195%	415%	150%	248%	33%	209%	67%

* The table reports the bias for alternative matching methods. The bias in the first column is $\hat{\beta}$. The estimator in the second column does not restrict matches to a common support region. The estimators in the third through seventh columns restrict matches to a common support region and the bias estimates correspond to $\hat{B}_{SP}$.

† The best predictor model given in the second footnote to Table 2, is used to estimate the probability of programme participation. The conditioning variables in the regression adjusted local linear models are site, race, age, education, previous training, the local unemployment rate, indicator variables for marital status and the presence of a child aged less than 6 in the household, and indicators for the quarter of the year and the year.

‡ A 5% trimming rule is used to determine the region of overlapping support (see Appendix C), and a fixed bandwidth equal to 0.06 and a biweight kernel, defined in Appendix A, are used for nonparametric estimates.

** The impacts in the table are mean monthly impacts for the six post-programme quarters, estimated using the experimental treatment and control data for the four JTPA training sites in our study. See the experimental impacts and the adjusted impacts in Table 4.